

Abdelmalek Essadi University*Algebra IV — Make-up Exam**Year: 2021***Exercise 1:**

Let $a \in \mathbb{R}$. Consider the matrix

$$A_a = \begin{pmatrix} 0 & 1 & a \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \in M_3(\mathbb{R}).$$

- 1) Determine the characteristic polynomial of the matrix $A_a + I_3$.
- 2) For which real values of a is the matrix A_a diagonalizable?
- 3) Write the minimal polynomial of A_a .
- 4) Choose $a = 1$ and set $A = A_1$.
 - 4-1) Determine a diagonalization relation of A .
 - 4-2) Let $n \in \mathbb{N}^*$. Consider the polynomial $P(X) \in \mathbb{R}[X]$ defined by $P(X) = (X + I)^n$. Determine the matrix $P(A)$.

Answer Area**Exercise 2:**

Consider the matrix $A \in M_3(\mathbb{R})$ defined by

$$A = \begin{pmatrix} -1 & 2 & 2 \\ -2 & 3 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Determine a Jordan reduction of the matrix A .

Answer Area**Exercise 3:**

Let $A \in M_n(\mathbb{C})$ be invertible and with n distinct eigenvalues. A square root of the matrix A is any matrix $M \in M_n(\mathbb{C})$ satisfying $M^2 = A$.

Show that the matrix A has 2^n square roots, all of which are polynomials in A .

Answer Area